Zoho Analytics Integration

1. Overview

1.1 Description

Zoho Analytics Integration connects **Zoho Analytics** (business intelligence and reporting platform) with **Newo Platform**.

It enables: - Verifying caller identity against a vendor database - Looking up invoice status by load number (due date, payment status, amount due, payee) - Verifying rate details by load number (rate, payee, payment amount) - Identifying caller's company/location with fuzzy name matching

This integration is designed for **logistics companies** who need **an AI phone agent to answer vendor (carrier/transporter) inquiries about invoices and rates** using data stored in Zoho Analytics tables.

1.2 Use Cases

- When a vendor calls and asks about invoice status -> Agent verifies identity, queries the invoices table, and reads back due date, payment status, amount due, and payee
- When a vendor calls to verify a rate -> Agent queries the rates table and reads back rate, payee, and payment amount
- When a caller needs to be verified -> Agent checks the vendors table by company name (and optionally phone number)
- When a caller mentions a company -> Agent identifies the correct location from the locations table, handling multiple matches with disambiguation

1.3 Supported Features

Feature	Supported	Notes
Invoice Status Lookup	[Yes]	By load number + vendor name
Rate Verification	[Yes]	By load number
Caller Identity Verification	[Yes]	By vendor name; phone number optional
Location Identification	[Yes]	LIKE matching for partial company names
Multi-Region Support	[Yes]	US, EU, IN, AU, JP, SA, CA data centers
OAuth2 Authentication	[Yes]	Standard redirect flow with token auto-refresh
Custom Canvas Scenarios	[Yes]	Logistics-specific scenarios and intent types
Booking/Scheduling	[No]	Not applicable - this is a data query integration
Write Operations	[No]	Read-only - agent never modifies Zoho Analytics data

2. Requirements

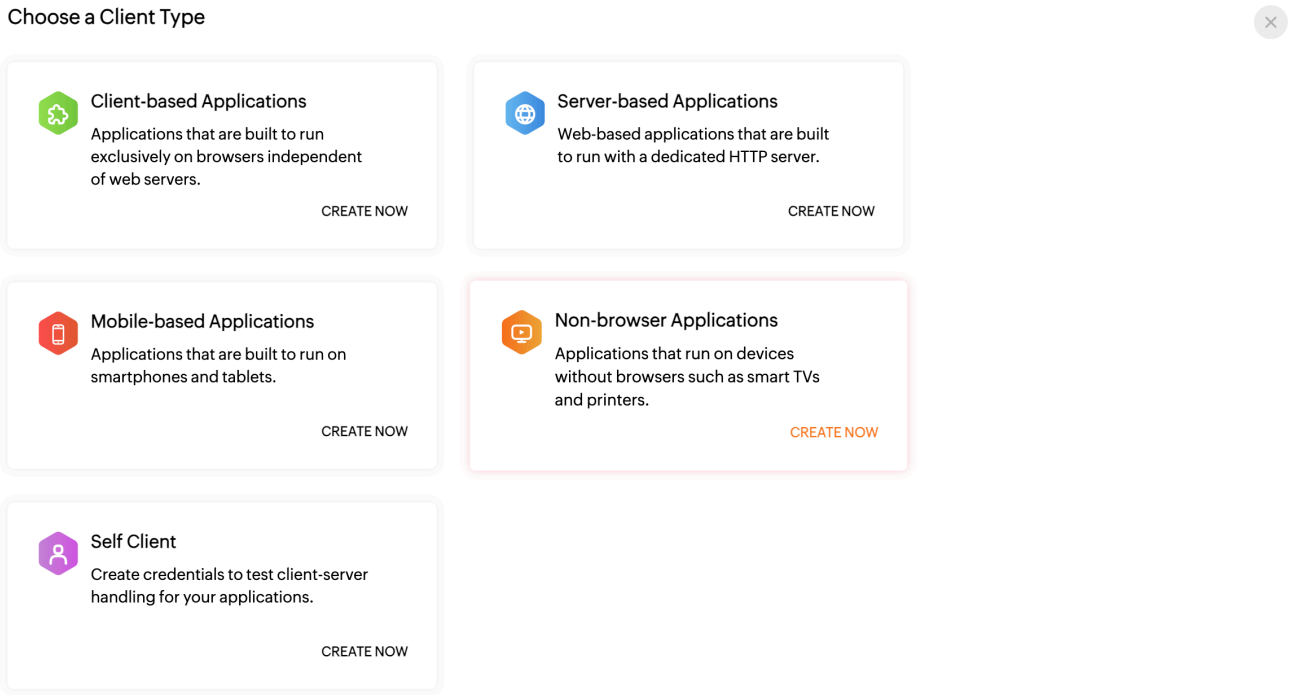
Before installation: - Active **Zoho Analytics** account with API access (free tier works) - **Server-based Applications** client registered in Zoho API Console - A Zoho Analytics workspace with 4 tables: **Invoices**,

Rates, **Vendors**, **Locations** - Data synced from your TMS (Transportation Management System) to these tables

3. How to Setup

3.1 Step 1 - Register OAuth Client in Zoho

1. Go to Zoho API Console
2. Click **Add Client** -> select **Server-based Applications**
3. Fill in:
4. **Client Name**: Newo Zoho Analytics
5. **Homepage URL**: https://newo.ai
6. **Authorized Redirect URIs**: https://static.newo.ai/oauth/index.html
7. Copy **Client ID** and **Client Secret**



3.2 Step 2 - Prepare Tables in Zoho Analytics

Create a workspace with 4 tables:

Invoices - invoice/payment data:

Column Name	Data Type
Load Number	**Plain Text**
Vendor Name	**Plain Text**
Due Date	**Date**
Payment Status	**Plain Text**
Amount Due	**Number**
Payee	**Plain Text**

Rates - rate and payment data:

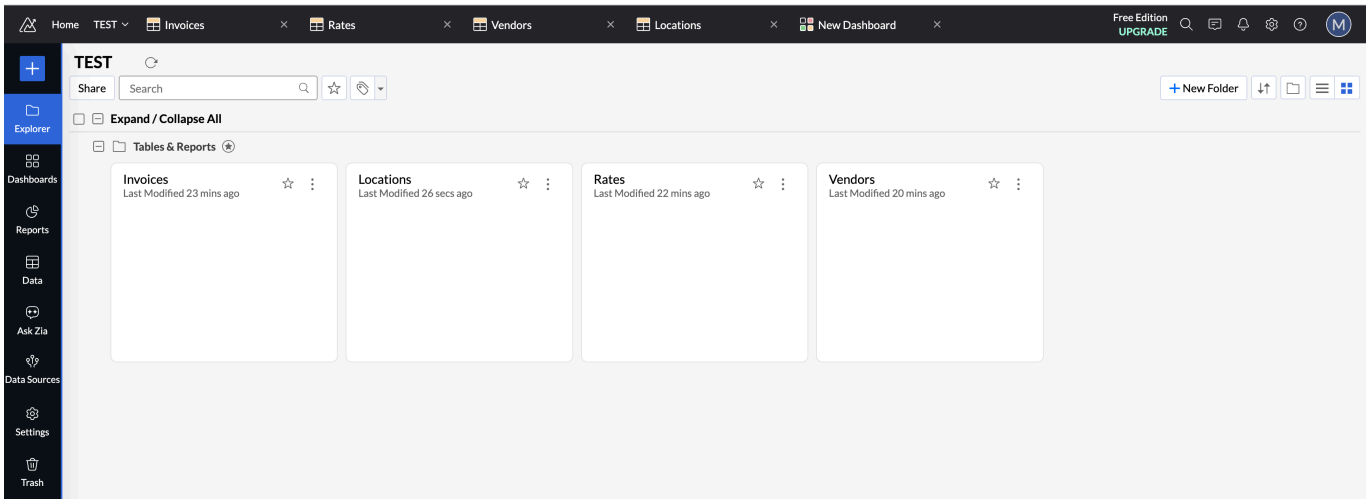
Column Name	Data Type
Load Number	**Plain Text**
Rate	**Number**
Payee	**Plain Text**
Payment Amount	**Number**

Vendors - vendor directory:

Column Name	Data Type
Vendor Name	**Plain Text**
Phone	**Plain Text**
Contact Person	**Plain Text**
Email	**Plain Text**

Locations - company/location directory:

Column Name	Data Type
Company Name	**Plain Text**
Location	**Plain Text**
Region	**Plain Text**
Account ID	**Plain Text**



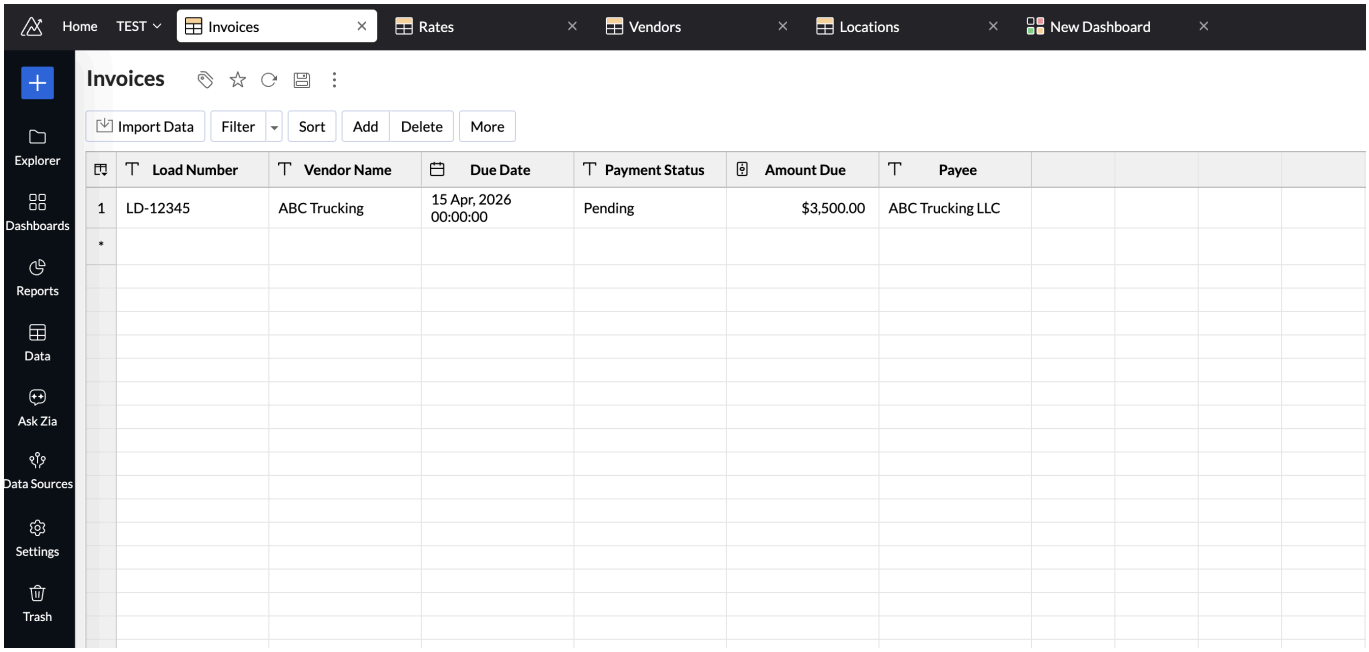
Important: Column names must match exactly (case-sensitive, with spaces). The integration queries by these exact column names.

3.3 Step 3 - Get Workspace ID and View IDs

Open each table in Zoho Analytics. The View ID is visible in the browser URL:

https://analytics.zoho.eu/workspace/WORKSPACE_ID/view/VIEW_ID

Note down: - **Workspace ID** - **View ID** for each of the 4 tables



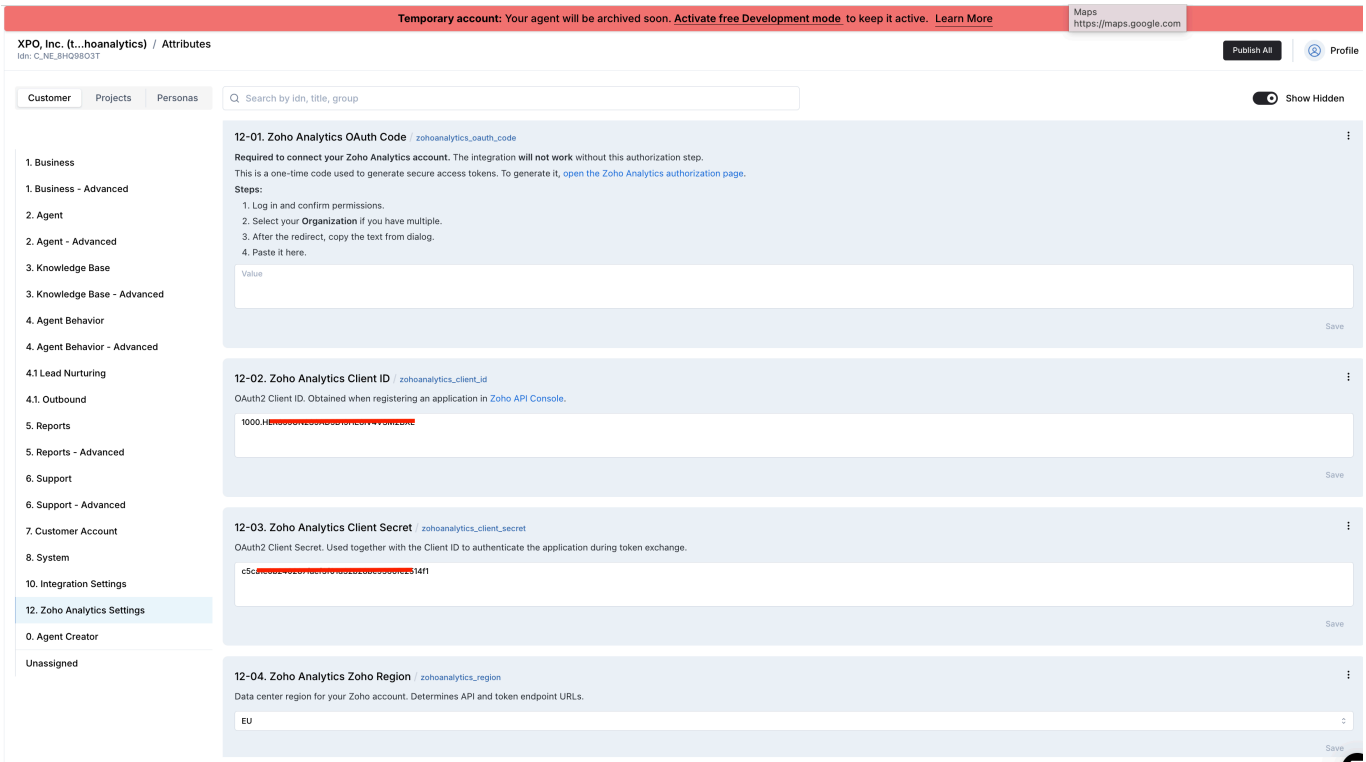
3.4 Step 4 - Get Organization ID

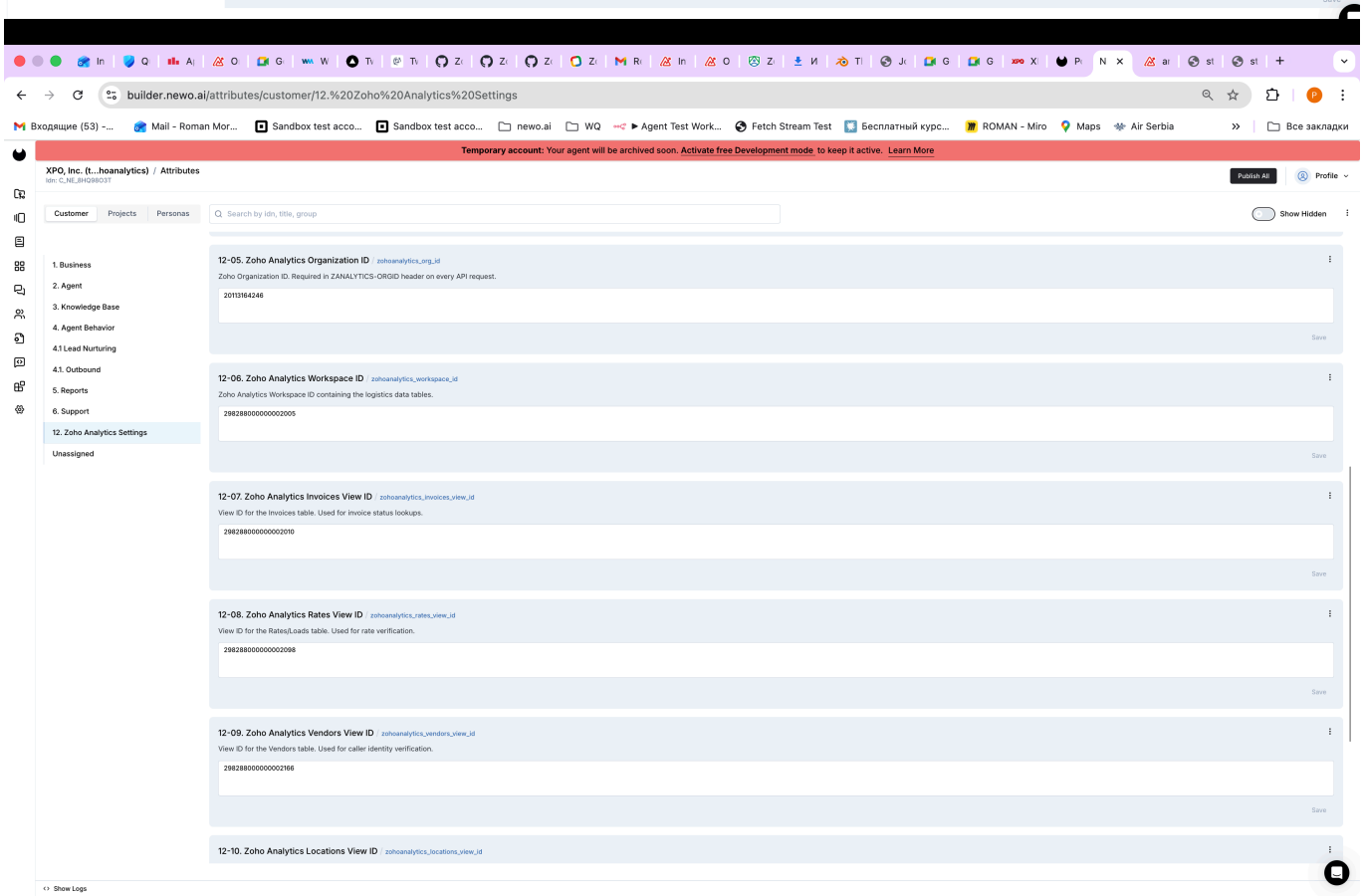
Go to **Zoho Analytics -> Setup -> General -> Organization Details**. Copy the **Organization ID** from the URL or the settings page.

3.5 Step 5 - Connect in Newo Platform

1. Open **builder.newo.ai -> Project -> Attributes**
2. Set the following attributes:

Attribute	Required	Description
zohoanalytics_client_id	[Yes]	OAuth2 Client ID from Zoho API Console
zohoanalytics_client_secret	[Yes]	OAuth2 Client Secret from Zoho API Console
zohoanalytics_region	[Yes]	Data center region: US, EU, IN, AU, JP, SA, or CA
zohoanalytics_org_id	[Yes]	Zoho Organization ID
zohoanalytics_workspace_id	[Yes]	Zoho Analytics Workspace ID
zohoanalytics_invoices_view_id	[Yes]	View ID for the Invoices table
zohoanalytics_rates_view_id	[Yes]	View ID for the Rates table
zohoanalytics_vendors_view_id	[Yes]	View ID for the Vendors table
zohoanalytics_locations_view_id	[No]	View ID for the Locations table
zohoanalytics_setup_scenarios	[No]	Auto-add logistics scenarios to canvas (default: True)
zohoanalytics_override_agent_attributes	[No]	Override SA attributes on publish (default: True)





1. **Fill in Client ID and Client Secret first**, then **Publish** the project
2. After publish, click the OAuth link in the **OAuth Code** attribute to authorize
3. Paste the code and **Publish** again

On publish, the SetupFlow automatically: - Resolves regional API and token endpoints based on selected region - Creates HTTP connectors for API requests and token refresh - Exchanges OAuth code for access + refresh tokens - Registers custom tools in ConvoAgent (verify caller, invoice lookup, rate verification, location identification) - Adds logistics-specific scenarios and intent types to the Workflow Builder Canvas

4. How to Use

4.1 How the Integration Works

The integration operates based on **Triggers and Actions** via the event-driven system.

- A **Trigger** is a code-phrase the agent says, detected by the ConvoAgent tool caller
- An **Action** is a query to Zoho Analytics Sync Export API

Available Triggers:

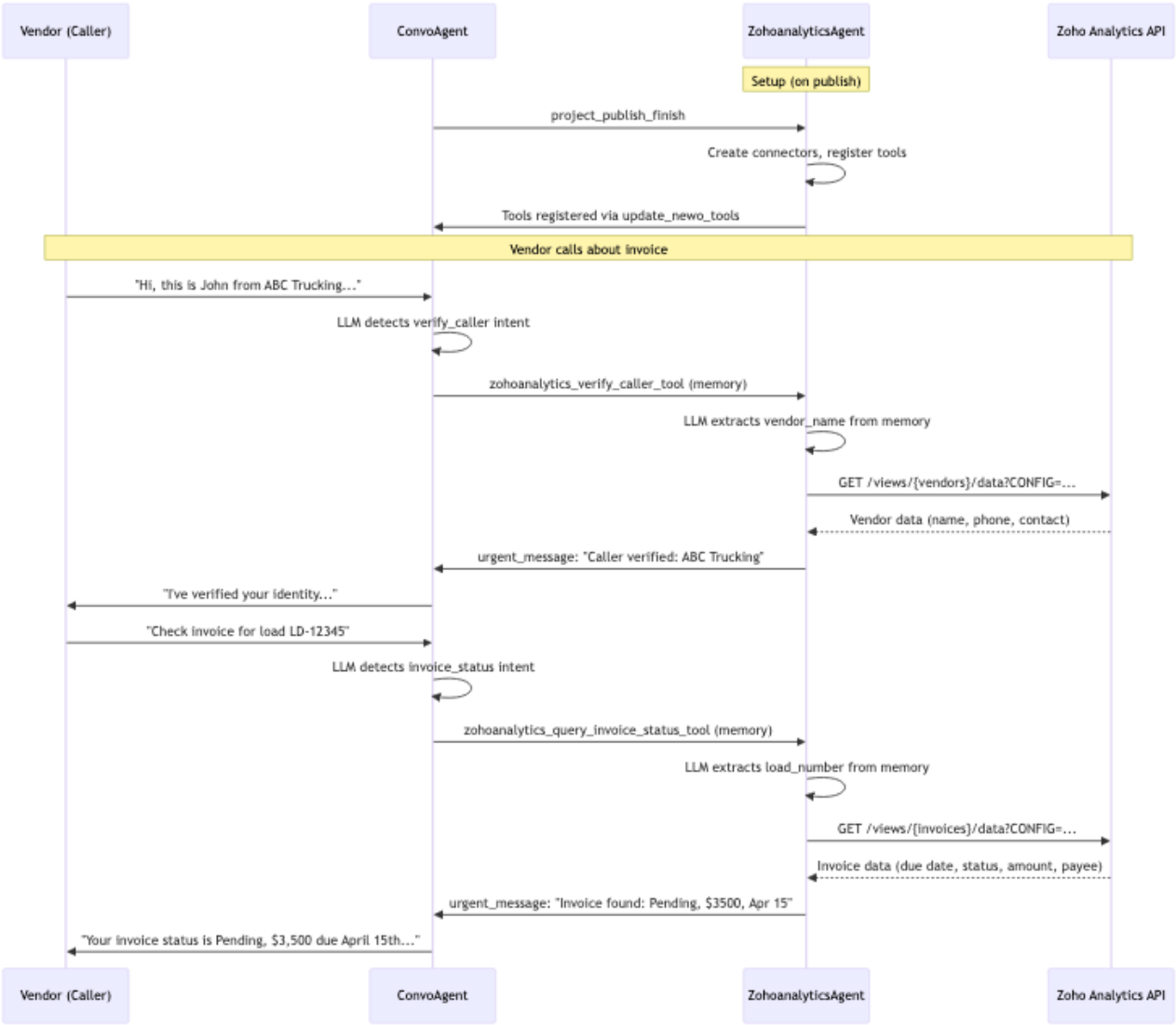
Trigger	Tool Event	Description
Verify Caller	zohoanalytics_verify_caller_tool	Agent says "Let me verify your information..."
Invoice Status Lookup	zohoanalytics_query_invoice_status_tool	Agent says "Let me look up the invoice..."
Rate Verification	zohoanalytics_query_rate_verification_tool	Agent says "Let me check the rate..."
Identify Location	zohoanalytics_identify_location_tool	Agent says "Let me look up that company..."

Available Actions:

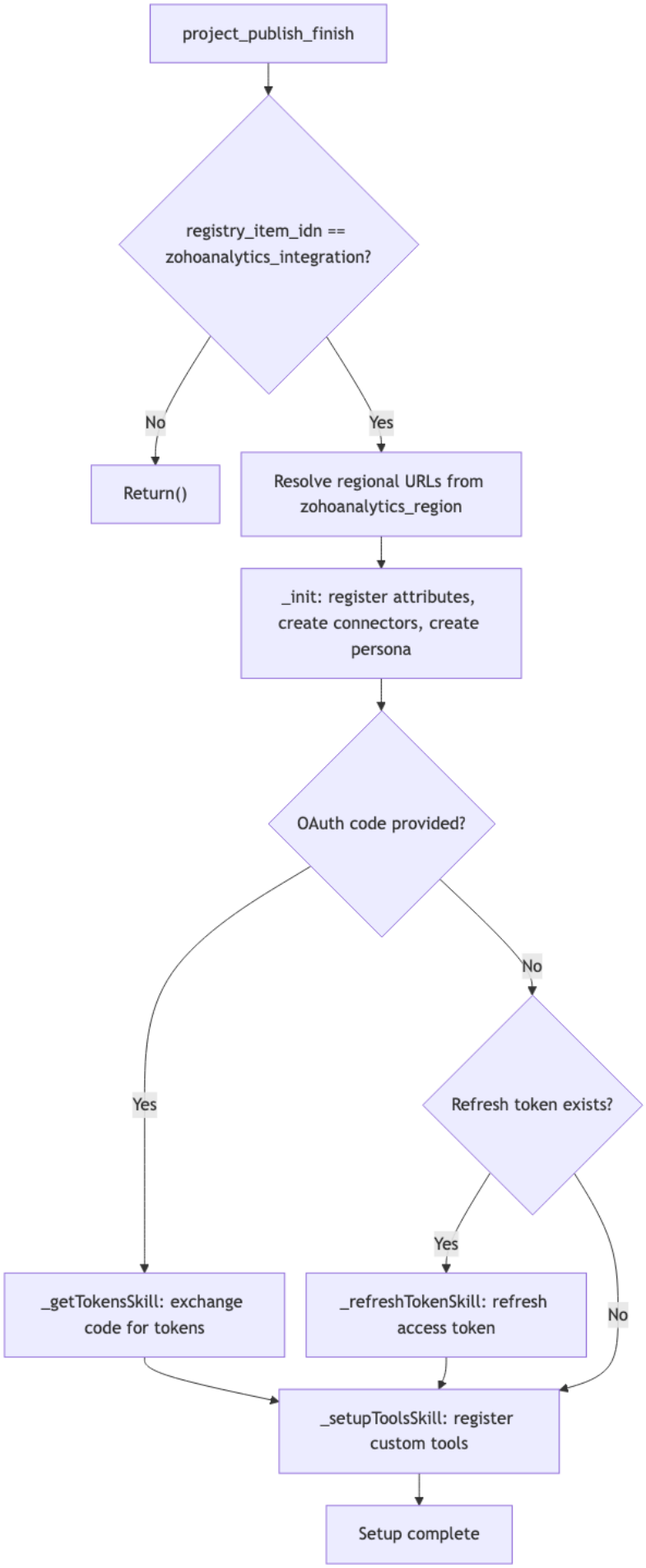
- **Verify Caller** - GET /workspaces/{id}/views/{vendors_view}/data with criteria "Vendor Name"='{name}'
- **Invoice Status** - GET /workspaces/{id}/views/{invoices_view}/data with criteria "Load Number"='{number}' AND "Vendor Name"='{name}'
- **Rate Verification** - GET /workspaces/{id}/views/{rates_view}/data with criteria "Load Number"='{number}'
- **Identify Location** - GET /workspaces/{id}/views/{locations_view}/data with criteria "Company Name" LIKE '%{name}%'

4.2 Architecture Diagrams

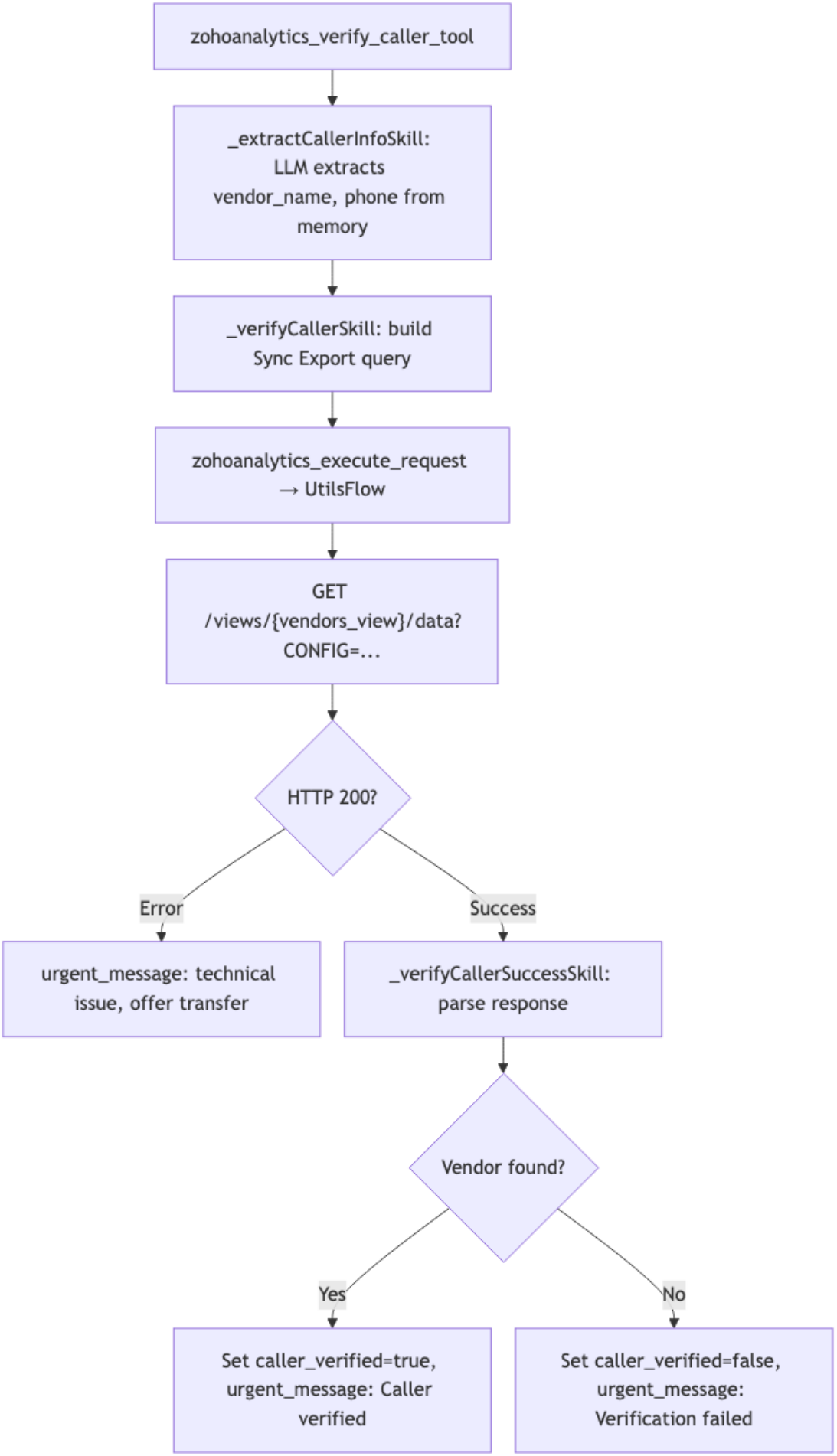
Overall Sequence:


Vendor (Caller)
ConvoAgent
ZohoanalyticsAgent
Zoho Analytics API
Setup (on publish)
project_publish_finish
Create connectors, register tools
Tools registered via update_newo_tools
Vendor calls about invoice
"Hi, this is John from ABC Trucking..."
LLM detects verify_caller intent
zohoanalytics_verify_caller_tool (memory)
LLM extracts vendor_name from memory
GET /views/{vendors}/data?CONFIG=...
Vendor data (name, phone, contact)
urgent_message: "Caller verified: ABC Trucking"
"I've verified your identity..."
"Check invoice for load LD-12345"
LLM detects invoice_status intent
zohoanalytics_query_invoice_status_tool (memory)
LLM extracts load_number from memory
GET /views/{invoices}/data?CONFIG=...
Invoice data (due date, status, amount, payee)
urgent_message: "Invoice found: Pending, $3500, Apr 15"
"Your invoice status is Pending, $3,500 due April 15th..."
Vendor (Caller)
ConvoAgent
ZohoanalyticsAgent
Zoho Analytics API

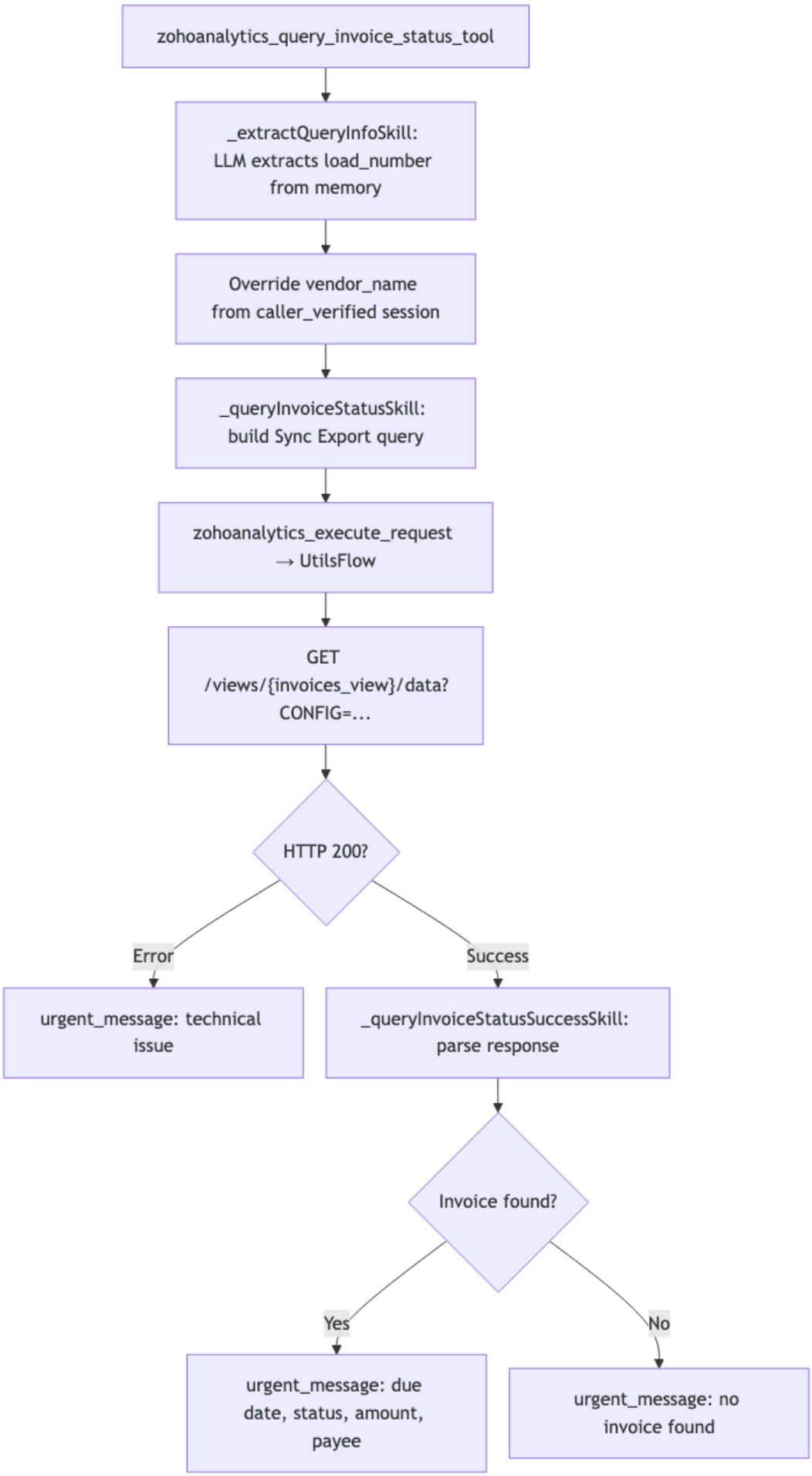
SetupFlow:



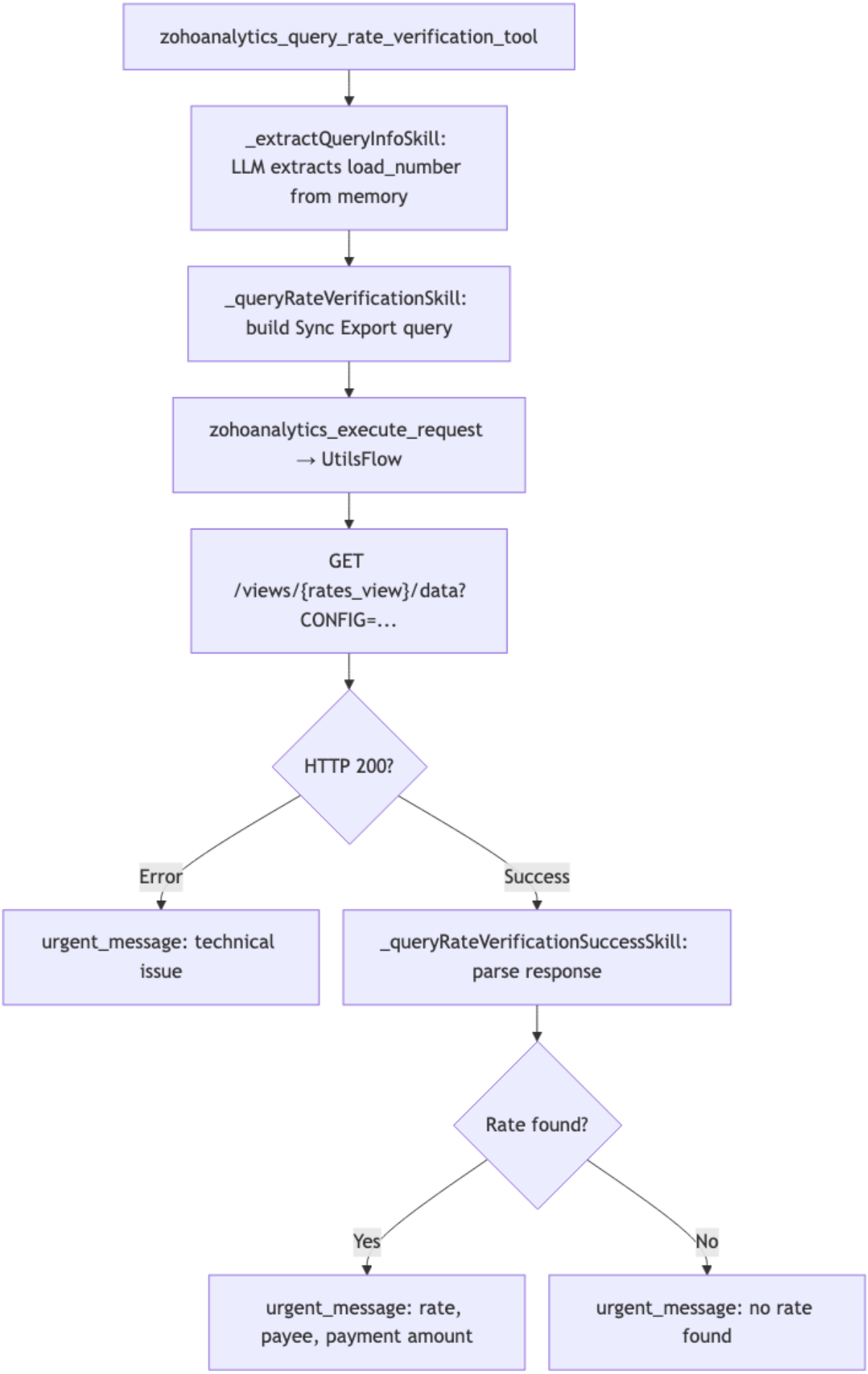
CallerVerificationFlow:



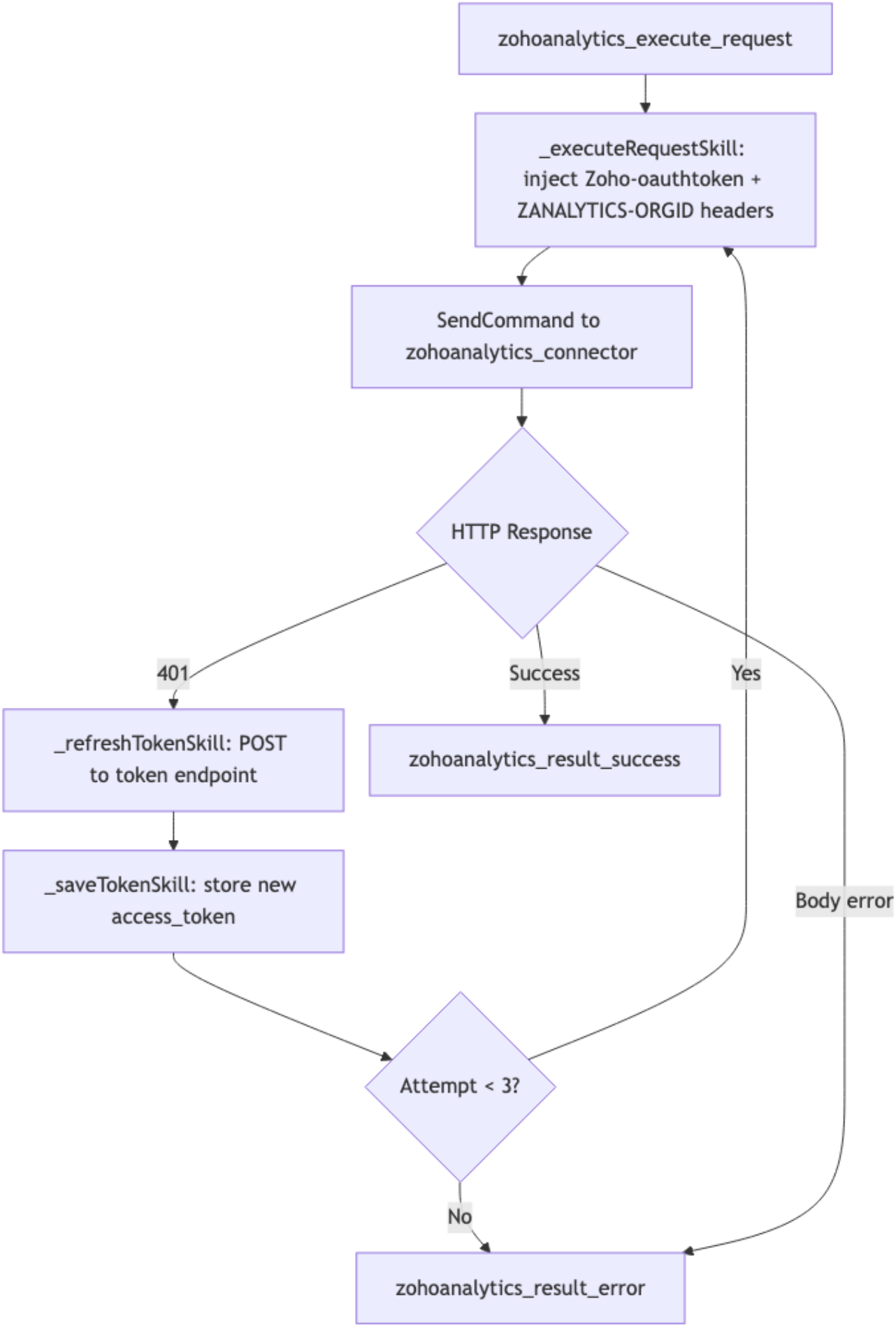
DataQueryFlow (Invoice Status):



DataQueryFlow (Rate Verification):


zohoanalytics_query_rate_verification_tool
_extractQueryInfoSkill: LLM extracts load_number from memory
_queryRateVerificationSkill: build Sync Export query
zohoanalytics_execute_request → UtilsFlow
GET /views/{rates_view}/data?CONFIG=...
HTTP 200?
Error
Success
urgent_message: technical issue
_queryRateVerificationSuccessSkill: parse response
Rate found?
Yes
No
urgent_message: rate, payee, payment amount
urgent_message: no rate found

UtilsFlow:



4.3 Where to Test

Testing can be done through the **Newo conversation interface** (chat or voice channel) by interacting with the agent as a vendor calling about invoices or rates.

4.4 Testing and Verification

To test the integration:

1. **Setup**: Publish the project and verify attributes appear in the builder (Client ID, Region, View IDs)
2. **OAuth**: Click the authorization link in OAuth Code attribute, authorize, paste code, publish again. Verify zohoanalytics_access_token is populated (hidden attribute)
3. **Caller Verification**: Say "Hi, this is [vendor name from Vendors table]" -> Agent should say "Let me verify your information" -> Confirm verification succeeded
4. **Invoice Lookup**: Ask "What's the status of my invoice for load [load number]?" -> Agent should return due date, payment status, amount, payee
5. **Rate Verification**: Ask "What's the rate for load [load number]?" -> Agent should return rate, payee, payment amount

If no action occurs:

- Ensure Client ID and Client Secret are set before first Publish
- Verify Region matches your Zoho account data center (check URL: .zoho.com = US, .zoho.eu = EU, etc.)
- Confirm all View IDs are filled (check for empty values)
- Re-publish after any attribute changes
- Check that table column names match exactly: Load Number, Vendor Name, Due Date, Payment Status, Amount Due, Payee, Rate, Payment Amount, Phone, Contact Person, Email, Company Name, Location, Region, Account ID
- Verify data exists in Zoho Analytics tables (empty tables return "not found")

Note: This integration performs **read-only** queries against Zoho Analytics. No data is modified in Zoho Analytics through the agent.

5. FAQ

Q: Does the integration modify data in Zoho Analytics? A: No. The integration is read-only - it only queries data via the Sync Export API. No rows are created, updated, or deleted.

Q: How does data get into Zoho Analytics tables? A: Data must be synced from your TMS (Transportation Management System) separately. Zoho Analytics supports imports from databases, CSV, cloud storage, and 250+ integrations. Data freshness depends on your sync frequency.

Q: What happens if the vendor is not found during verification? A: The agent informs the caller that verification failed and offers to transfer to a human representative.

Q: Can the agent handle multiple invoice lookups in one call? A: Yes. After answering an invoice question, the agent asks "Is there anything else?" The caller can ask about another load number.

Q: What if the table has more than 1 million rows? A: The Sync Export API only works for tables under 1 million rows. For larger tables, an async export approach would be needed (not currently implemented).

Q: Which Zoho regions are supported? A: US, EU, India, Australia, Japan, Saudi Arabia, and Canada. Set the correct region in the zohoanalytics_region attribute.

Q: Can I use Zoho Analytics free tier? A: Yes. The free tier (2 users, 5 workspaces, 10,000 rows) is sufficient for testing and small deployments.